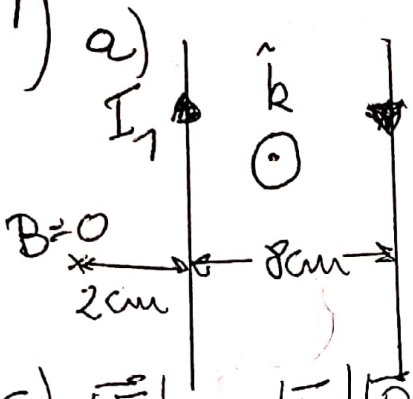
1) a)



$0 = \frac{\mu_0}{2\pi}\left(\frac{I_1}{2cm} - \frac{I_2}{10cm}\right) \rightarrow I_2 = 5I_1 \rightarrow \boxed{I_2 = 5\,A}$

b) $\vec{B} = -\hat{k}\frac{\mu_0}{2\pi}\left(\frac{I_1}{0{,}04m} + \frac{I_2}{0{,}04m}\right) \rightarrow \boxed{\vec{B} = -30\,\mu T\,\hat{k}}$

c) $|\vec{F}| = q|\vec{v}||\vec{B}| \rightarrow \boxed{|\vec{F}| = 7{,}2\,\mu N}$

2) a) VALE LA APROX. DE SOLENOIDE INFINITO

$L = \mu_0 \frac{N^2}{l}\pi R^2 \rightarrow \boxed{L = 23{,}69\,mH}$ (APUNTE INDUCTANCIA HOJA 2)

b) $M = \mu_0 N_1 N_2 \pi R_1^2 / l \rightarrow \boxed{M = 15{,}79\,mH}$ (APUNTE INDUCTANCIA HOJA 9)

3) $|\vec{B}_0| = \frac{\mu_0 I R^2}{2(R^2+D^2)^{3/2}} \rightarrow (R^2+D^2)^{3/2} = \frac{\mu_0 I R^2}{2|\vec{B}_0|} \rightarrow$

$\rightarrow D^2 = \left(\frac{\mu_0 I R^2}{2|\vec{B}|_0}\right)^{2/3} - R^2 \rightarrow D = 10\,cm$. LUEGO:

$|\vec{B}| = \frac{\mu_0 I R^2}{2[R^2+(3D)^2]^{3/2}} \rightarrow \boxed{|\vec{B}| = 0{,}01579\,\mu T}$

4) a) $120V = X_L \cdot 3A \rightarrow X_L = 40\,\Omega \rightarrow 160\pi\frac{1}{5}L = 40\,\Omega \rightarrow$

$\rightarrow \boxed{L = \frac{1}{4\pi}H = 79{,}6\,mH}$

$X_C = \frac{1}{160\pi\frac{1}{5}C} = 33{,}16\,\Omega < X_L \rightarrow$ INDUCTIVO

b) $Z = \sqrt{R^2+(X_L-X_C)^2} = 12{,}12\,\Omega$ $\cos\phi = \frac{R}{Z} \rightarrow \boxed{\cos\phi = 0{,}825}$

$f_R = \frac{1}{2\pi\sqrt{LC}} \rightarrow \boxed{f_R = 72{,}83\,Hz}$

c) $X_L - X_C = X_C' - X_L \rightarrow X_C' = 2X_L - X_C = 4684\,\Omega = \frac{1}{160\pi\frac{1}{5}C'}$

$\rightarrow \boxed{C' = 42{,}5\,\mu F}$

Escaneado con CamScanner